

Forecasting National CO₂ Emissions Three Years Ahead

A Regression Study of 17 South and Southeast Asian Economies (1990–2024)

Programming in Python — Final-Term Project Report
Summer 25-26 Semester

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[Section C]

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1. Project Title and Group Information

Project title: Forecasting National CO₂ Emissions Three Years Ahead — A Regression Study of South and Southeast Asian Economies.

This report documents an end-to-end regression workflow built on the Our World in Data (OWID) CO₂ and Greenhouse Gas Emissions dataset. The notebook and this report were prepared jointly by the group members listed on the title page; individual contributions are itemized in Section 12.

Field	Detail
Course	Programming in Python — Final-Term Project
Semester	Summer 25-26
Pathway	Regression
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Submission date	[12/09/26]

2. Problem Statement, Stakeholders, Objective, and Success Criteria

2.1 Problem statement

Governments, development banks, and climate-policy analysts need forward-looking estimates of national CO₂ output to plan energy transitions, negotiate emissions targets, and allocate climate financing. This project formulates a regression problem: given a country's energy, demographic, and emissions history up to a given year, predict that country's total CO₂ emissions three years into the future.

2.2 Unit of analysis, inputs, and target

- Unit of analysis: one country-year observation (17 countries × 35 years).
- Inputs: current-year population, GDP, primary energy consumption, energy per capita, CO₂ per capita, methane, nitrous oxide, current CO₂, and 1-, 2-, and 3-year lags of CO₂, population, GDP, and primary energy consumption, plus a one-hot encoded country identifier.
- Target: log-transformed total CO₂ emissions (million tonnes) for the same country, three years after the observation year.

2.3 Stakeholders

- National energy and environment ministries planning emissions-reduction pathways.
- Regional development organizations (e.g., ADB, World Bank) assessing climate-finance needs.
- Researchers and NGOs tracking progress toward international climate commitments (e.g., NDCs).

2.4 Objective and success criteria

The objective is a reproducible, leakage-safe pipeline that outperforms a naive baseline by a wide margin on untouched test data while remaining interpretable enough to defend to a non-technical audience. A useful result means the model's test-set error is small relative to the natural scale of national emissions (tens to thousands of megatonnes) and that its largest errors are understood and explained rather than hidden. Concretely, success was

defined as: (1) beating a mean-value baseline by at least an order of magnitude in RMSE, and (2) achieving a test R^2 above 0.85 while keeping every preprocessing step confined to information available at prediction time.

3. Dataset Source, License, Data Dictionary, Privacy, and Ethical Considerations

3.1 Source and provenance

The raw data is Our World in Data's `owid-co2-data.csv`, a compiled panel combining the Global Carbon Project, the Energy Institute Statistical Review of World Energy, UN population statistics, and the World Bank, distributed under Creative Commons BY 4.0. The raw file contains 50,411 rows and 79 columns spanning the years back to 1750 and includes both individual countries and aggregate/regional rows (continents, income groups, world totals).

3.2 Country panel and scope

The analysis restricts the raw data to a deliberately chosen 17-country panel — 8 South Asian countries (Bangladesh, Pakistan, Nepal, Sri Lanka, India, Bhutan, Afghanistan, Maldives), 8 Southeast Asian countries (Vietnam, Myanmar, Cambodia, Laos, Philippines, Indonesia, Thailand, Malaysia), and Mongolia — over 1990–2024. All 17 countries were confirmed present in the raw OWID data with no missing entries.

3.3 Data dictionary (key variables used)

Variable	Type / Unit	Description
country / iso_code	string	Country name and ISO-3 code; panel key.
year	integer	Calendar year, 1990–2024.
co2	float, million tonnes	Total annual CO ₂ emissions — the modelling target's basis (log, 3-year-ahead).
population	float	Mid-year national population.
gdp	float, current international \$	Gross domestic product.
primary_energy_consumption	float, TWh	Total primary energy use.
energy_per_capita	float, kWh/person	Per-capita energy consumption.
co2_per_capita	float, tonnes/person	Per-capita CO ₂ emissions.
methane / nitrous_oxide	float, Mt CO ₂ -eq	Non-CO ₂ greenhouse-gas emissions.
co2_growth_abs	float	Excluded from features — identified as a direct linear transform of <code>co2</code> (see Section 5).

3.4 Privacy and ethics

All records are national-level, publicly aggregated statistics; no personal or individually identifiable data is present, so there are no personal-privacy concerns. Ethically relevant limitations include: (1) representation gaps — emissions data quality is historically weaker for lower-income and conflict-affected countries in the panel (e.g., Afghanistan), so estimates for those countries carry more uncertainty than for data-rich countries like India; (2) responsible-use boundary — the model should not be used to justify withholding climate finance from a country based on a projected emissions increase, since projections reflect historical growth trajectories, not policy intent; and (3) the dataset is entirely secondary and observational, so no causal claims about drivers of emissions should be drawn from it.

4. Data Audit, Exploratory Analysis, and Visualization Findings

4.1 Removing aggregates and building a complete panel

Rows without an ISO country code (continents, income groups, world totals) were dropped first, removing 7,931 aggregate/regional rows from the raw 50,411 ($\rightarrow$ 42,480 country-level rows). Filtering to the 17-country panel and 1990–2024 produced exactly the expected 595 rows (17 countries $\times$ 35 years), confirmed against a fully-crossed country $\times$ year grid with zero missing (country, year) combinations and zero rows where the target variable co2 itself was missing.

4.2 Duplicate and label-consistency checks

No duplicate (country, year) rows were found. Every country mapped to exactly one ISO code, and a case/whitespace-normalized comparison of country name strings found zero hidden name variants — the country column is clean and requires no relabeling.

4.3 Missing values

Within the 595-row panel, 34 of the original 79 columns had at least one missing value, ranging from 2 missing entries (cement_co2 family) up to all 595 rows missing (fully absent columns such as share_global_other_co2 and other_industry_co2). Columns missing more than 25% of their values were dropped entirely (15 columns removed, leaving 65), and the remaining numeric gaps were filled with the training-consistent column median. No missing values remained in the 11 features ultimately selected for modelling (Section 5).

4.4 Range and outlier checks

- Non-positive CO2 values: 0 rows (required for the later log transform to be valid).
- Out-of-range checks on population, primary_energy_consumption, and energy_per_capita: 0 violations.
- Per-country IQR outlier screening on CO2 (rather than a single panel-wide threshold, since India's scale would otherwise mask genuine anomalies in smaller economies) flagged 7 rows: five consecutive high years in Cambodia (2020–2024) and two in Nepal (2023–2024). These were reviewed and retained as real, sustained increases in national emissions rather than data errors, since they persist across multiple years rather than appearing as single-point spikes.

4.5 Distribution and relationship findings

The panel's CO2 values are heavily right-skewed (mean 170.4 Mt, median 21.9 Mt) because India's emissions are an order of magnitude larger than most of the panel; a log transform of CO2 produces a much more symmetric distribution, motivating the log-target modelling choice used throughout. A population-vs-CO2 scatter plot on log-log axes shows a broadly positive relationship, but with visible scatter — economies with similar population sizes still differ substantially in emissions, indicating that population alone would be an inadequate predictor and additional energy/economic features are needed.

5. Split Strategy, Preprocessing Decisions, and Leakage Controls

5.1 Feature selection and a leakage check

Before selecting features, the pipeline explicitly tested whether the candidate column co2_growth_abs could be reconstructed from the target itself. Recomputing each country's year-over-year CO2 difference reproduced co2_growth_abs exactly, confirming it is a deterministic transform of the target and would leak target information if used as a predictor. It was therefore excluded. The final 11 retained columns were: country, year, iso_code, co2, population, gdp, primary_energy_consumption, energy_per_capita, co2_per_capita, methane, and nitrous_oxide.

5.2 Feature engineering

The target was log-transformed ($\log_co2 = \ln(co2 + 1e-6)$) to correct the right-skew identified in Section 4.5. One-, two-, and three-year lag features were then constructed within each country for `co2`, `population`, `gdp`, and `primary_energy_consumption`, giving the model historical trend information without looking ahead. Rows without enough history to compute all lags were dropped, leaving 24 engineered columns. The forecasting target, `target_3yr_ahed`, was created by shifting each country's `log_co2` series three years backward (i.e., attaching the future value to the current row); rows without a valid 3-year-ahead outcome (the final three years of each country's series) were dropped from the modelling set.

5.3 Chronological train / validation / test split

Because the data is a time-ordered panel and the target is itself a future value, the split is chronological rather than randomized, which prevents the model from ever training on a period that is later than what it is tested on:

Split	Base years	Rows	Purpose
Train	1993–2015	391	Fit models and search hyperparameters.
Validation	2016–2019	68	Select the best hyperparameters and the winning model family.
Test	2020–2021	34	Single, untouched final evaluation (Section 8).

5.4 Encoding, scaling, and leakage control

The categorical country column was one-hot encoded (`drop_first=True`) separately within each split after the split boundary was fixed, producing 37 model-ready features. All continuous features were standardized with scikit-learn's `StandardScaler` fitted only on the training split; the validation and test splits were transformed using that same fitted scaler and never used to (re)fit it. This mirrors the course's leakage-control requirement (FR-2): no test or validation information influences imputation, scaling, or encoding decisions.

6. Baseline Definition and Result

The baseline is the simplest defensible reference for a regression task: predict the training set's mean log-CO2 value for every validation row, regardless of country or year. Its error is reported on the original (exponentiated) CO2 scale so it is directly comparable to the final model's test error in Section 8.

Metric	Mean Baseline (Validation Set)
MAE	268.03 Mt
RMSE	666.36 Mt
R ²	−0.18

The negative R² confirms the baseline is worse than simply predicting the mean for every observation would be if the mean were computed on the validation set itself — an expected outcome given the wide spread between small (e.g., Bhutan, Maldives) and large (India) economies in the panel, and it sets a very low bar that any credible model must clear by a wide margin.

7. Candidate Models, Model-Choice Reasoning, Hyperparameters, and Validation Method

7.1 Models considered

Four regressors were trained and compared on identical training data and evaluated with identical validation data, satisfying the fair-comparison requirement: K-Nearest Neighbours, Random Forest, XGBoost, and Support Vector Regression (SVR). This spans a simple distance-based method, two tree ensembles of differing bias/variance character, and a margin-based linear/kernel method — a defensible spread given the small sample size (391 training rows) and moderate feature count (37).

7.2 Hyperparameter search (grid search, validated on the held-out validation split)

Model	Grid Searched	Best Configuration	Best Validation RMSE (log scale)
K-Nearest Neighbours	n_neighbors $\in$ {3,5,7,10}; weights $\in$ {uniform, distance}	n_neighbors=3, weights=distance	0.2192
Random Forest	n_estimators $\in$ {100,200}; max_depth $\in$ {5,10,None}	n_estimators=200, max_depth=10	0.2264
XGBoost	n_estimators $\in$ {100,200}; max_depth $\in$ {3,5}; learning_rate $\in$ {0.01,0.1}	n_estimators=100, max_depth=5, learning_rate=0.1	0.2266
SVR	C $\in$ {0.1,1,10}; kernel $\in$ {linear, rbf}	C=1, kernel=linear	0.2075

Tuning used only the training and validation splits — the test split was never inspected during model or hyperparameter selection, consistent with the no-test-set-tuning requirement. random_state=42 was fixed for the stochastic learners (Random Forest, XGBoost) so the search is exactly reproducible.

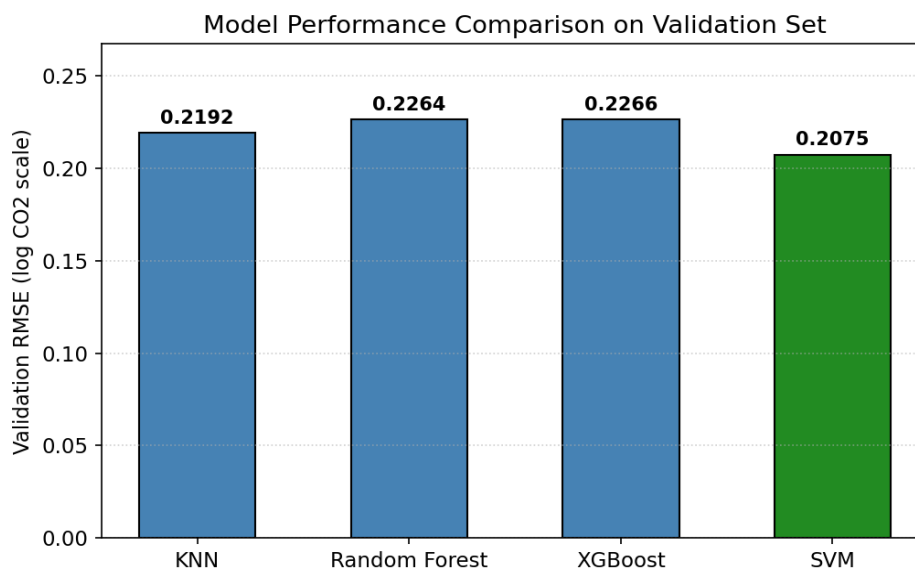


Figure 1. Validation RMSE (log-CO₂ scale) for the best configuration of each candidate model. Linear-kernel SVR was the winner.

7.3 Model-choice reasoning and defense

The linear-kernel SVR achieved the lowest validation RMSE (0.2075), narrowly ahead of KNN (0.2192) and the two tree ensembles (≈ 0.226). This is a plausible outcome given the setup: after log-transforming the target and standardizing all inputs, the dominant signal is close to linear — current CO₂ and its own recent lags are the strongest predictors of the near-future value (Section 7.4), and a linear-kernel SVR is well suited to a small,

standardized, largely linear-relationship dataset (391 training rows, 37 features) where more flexible models (RBF-SVR, deep trees) are more prone to overfitting. The RBF-kernel SVR variant performed far worse (RMSE up to 0.99 at $C=0.1$), reinforcing that the underlying relationship in this feature space is closer to linear than nonlinear. SVR requires standardized inputs (already satisfied by the shared preprocessing pipeline), has a single regularization parameter (C) that was tuned via the validation grid, and trades some interpretability (no direct feature-importance output, addressed by reporting tree-based importances as a proxy in Section 7.4) for stronger generalization on this dataset size. It is therefore preferable to the tree ensembles here specifically because of the small-sample, largely-linear regime — not as a general claim that SVR dominates tree methods.

7.4 Feature importance (tree-based models, used as an interpretability proxy)

Because SVR does not expose native feature importances, the Random Forest and XGBoost fits are used to interpret which inputs drive the forecast, with the caveat that this describes those models' behavior rather than SVR's coefficients directly.

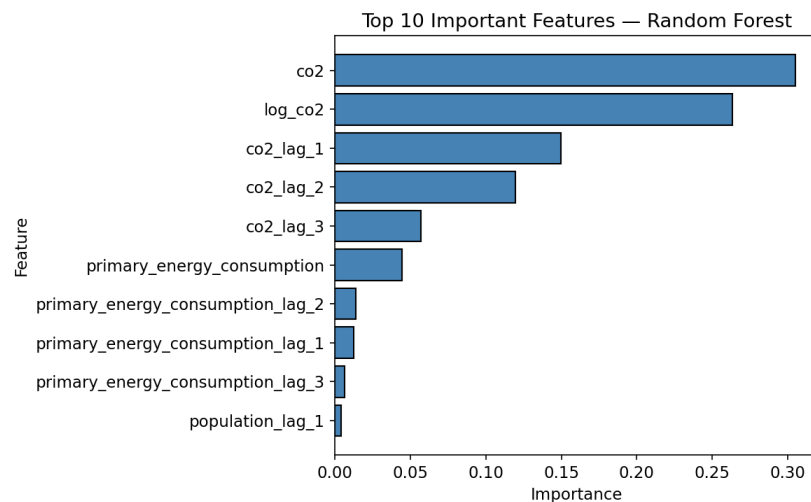


Figure 2. Random Forest top-10 feature importances.

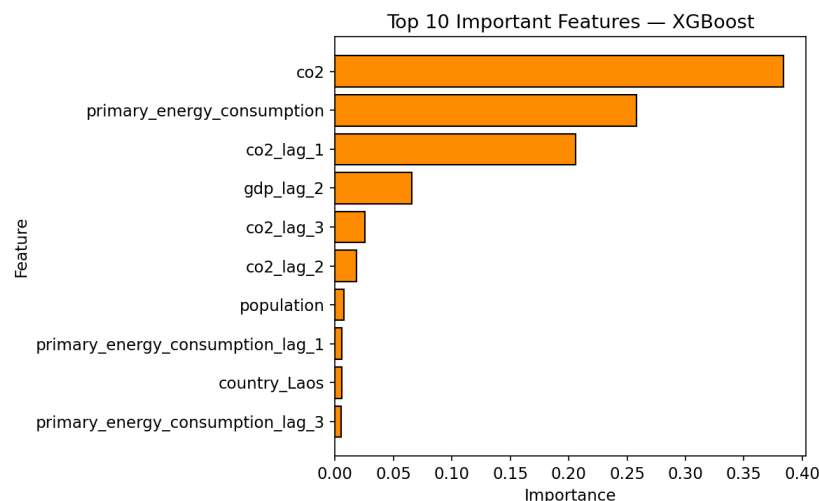


Figure 3. XGBoost top-10 feature importances.

Both models agree that the current-year co2 value and its recent lags (co2_lag_1/2/3) dominate the prediction, followed by primary_energy_consumption and its lags — an intuitive result for a 3-year-ahead autoregressive

forecast: recent history is the strongest available signal, and energy consumption is the strongest structural driver behind it.

8. Final Metrics, Diagnostic Plots, Model Comparison, and Computational Cost

8.1 Final test-set evaluation

The winning model (SVR, C=1, linear kernel) was evaluated exactly once on the untouched 2020–2021 test split (34 rows), with predictions exponentiated back to the original CO₂ (million tonnes) scale for interpretability:

Metric	Value
Test MAE	58.34 Mt
Test RMSE	194.20 Mt
Test MAPE	14.76%
Test R ²	0.9289
Test RMSE (log scale)	0.1823
Test MAE (log scale)	0.1381

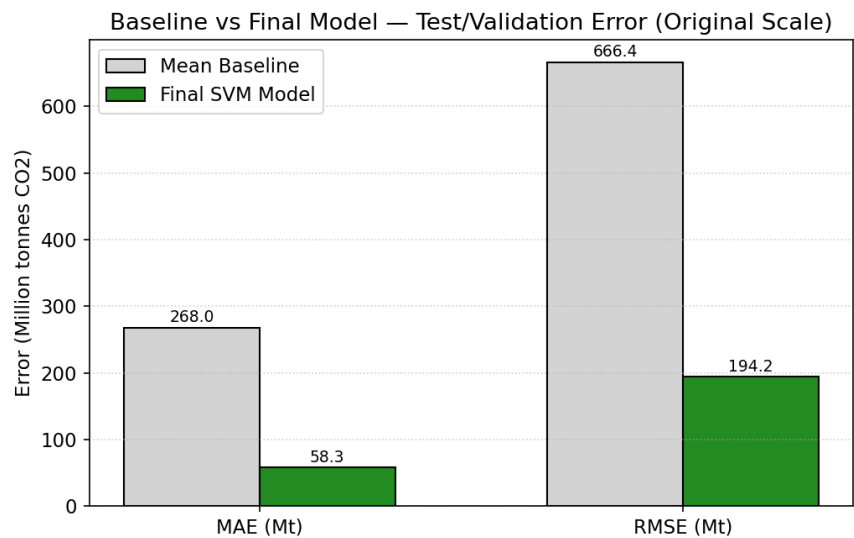


Figure 4. Baseline vs. final model error on the original CO₂ scale (note: baseline is on the validation split, final model on the test split, as defined by the workflow).

Against the mean baseline (RMSE 666.36 Mt, R² −0.18), the final model reduces RMSE by roughly 71% and moves R² from negative to 0.93 — comfortably clearing the success criteria set in Section 2.4.

8.2 Diagnostic plots

Residuals (actual minus predicted log-CO₂) were plotted against predicted values and as a histogram overlaid with a fitted normal curve. The residuals-vs-predicted plot showed no strong funnel or curvature pattern, and the residual histogram was reasonably close to the overlaid normal curve, supporting the log transform's role in stabilizing variance across the wide range of country sizes in the panel.

8.3 Computational cost

All four candidate models trained in well under a second per configuration on this dataset size (391 training rows), so wall-clock training time was not a differentiating factor in model choice; the grid searches for KNN, Random Forest, XGBoost, and SVR together completed in a few seconds on a standard Colab CPU runtime. This is expected given the small panel size and is noted here to satisfy the computational-documentation requirement, not because compute cost drove the model decision.

8.4 Out-of-sample forecast (2025–2027)

After model selection, the winning SVR configuration was refit on the entire historical dataset (493 rows spanning all available years) using a scaler refit on that full dataset, and used to forecast total panel CO₂ for 2025–2027 with a 95% interval derived from the residual standard deviation on the log scale.

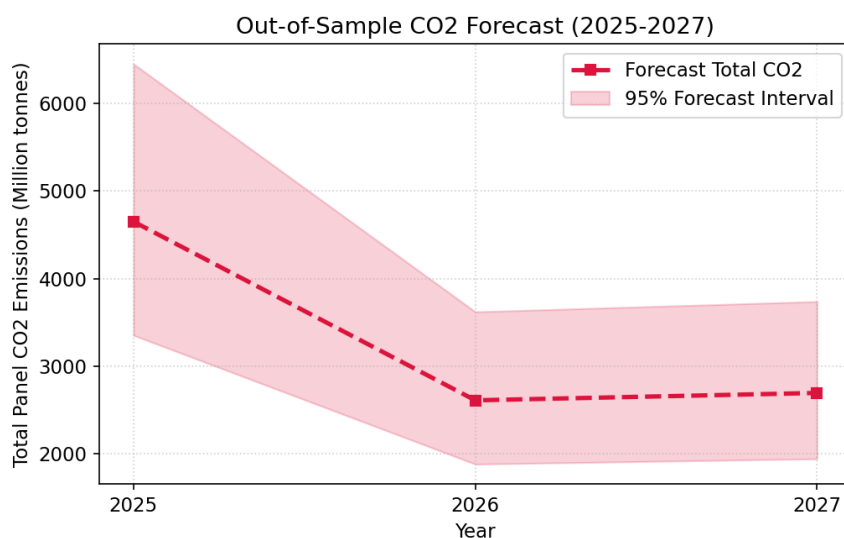


Figure 5. Forecast of total panel CO₂ emissions, 2025–2027, with a 95% interval.

Year	Forecast (Mt)	Lower 95% (Mt)	Upper 95% (Mt)
2025	4,651.6	3,353.6	6,451.8
2026	2,611.3	1,882.7	3,622.0
2027	2,695.7	1,943.5	3,738.9

The 2025 point forecast is notably higher than 2026–2027 and its interval is wide, which is discussed as a limitation in Section 9 rather than presented as a confirmed trend reversal.

9. Error Analysis, Imbalance, Assumptions, Bias, Uncertainty, and Generalization Limitations

9.1 Largest test-set errors

The ten largest absolute test errors are dominated by India and, to a lesser extent, Pakistan and Indonesia:

Country	Base Year	Actual CO2 (Mt)	Predicted CO2 (Mt)	Abs. Error (Mt)	% Error
India	2020	3,062.8	4,092.9	1,030.1	33.6%
India	2021	3,193.5	3,632.1	438.6	13.7%
Pakistan	2021	179.8	284.2	104.4	58.0%
Indonesia	2020	762.4	835.5	73.2	9.6%
Pakistan	2020	186.9	258.1	71.2	38.1%

Two distinct error patterns emerge. First, India's absolute errors are the largest in megatonnes because it is by far the largest emitter in the panel — even a well-fitting model produces bigger absolute misses on the largest scale. Second, Pakistan's percentage errors are the largest proportionally (up to 58%), which is more concerning: it suggests the model is comparatively less reliable for mid-sized, faster-growing economies than for very large or very small ones. This is consistent with the panel's scale imbalance (India's emissions are roughly two orders of magnitude above the smallest countries in the panel) rather than a class-imbalance issue, since this is a regression task.

9.2 Assumptions

- The 3-year-ahead relationship between recent history and future emissions is assumed to be reasonably stable over time; the model has not been tested through a genuine structural break (e.g., a global energy-price shock or pandemic-scale disruption) inside the test window.
- One-hot encoding a fixed 17-country panel means the model cannot generalize to a country outside this list without retraining.
- The $\log(\text{CO}_2 + 1\text{e-}6)$ transform assumes strictly positive emissions, verified in Section 4.4, and treats multiplicative (percentage) errors as roughly homoscedastic on the log scale.

9.3 Bias and representation risks

Because parameters were tuned to minimize validation RMSE across all 17 countries pooled together, the model implicitly weights toward patterns that are common across the whole panel; countries with more volatile or data-sparse histories (e.g., Afghanistan) are more likely to be under-served by a pooled model than a country-specific one would be. The Section 9.1 finding that Pakistan's percentage error is highest reinforces that pooled, scale-dominated training can favor the largest economies' patterns.

9.4 Uncertainty and its limits

The reported 95% forecast interval is derived from a single residual-standard-deviation estimate on the 2020–2021 test residuals and assumes those residuals are representative of 2025–2027 forecast error — an assumption that weakens the further the forecast horizon moves from the test period, which is one reason the 2025 interval in Figure 5 is wide. Because the model was tuned on a small validation set (68 rows) and evaluated on an even smaller test set (34 rows), single point-estimates for RMSE/MAE should be read with caution; repeated-seed or cross-validation-based uncertainty estimates were not computed for the tree models beyond the single train/validation split used for tuning, which is a limitation relative to the ideal of reporting variability across folds or seeds.

9.5 Generalization limits and responsible-use boundary

This model should not be used to (1) make country-specific policy commitments without country-level review, given the Pakistan-level percentage errors observed; (2) forecast beyond a 3-year horizon, since it was trained and validated exclusively for that horizon; or (3) forecast for countries outside the 17-country panel. It is best used as a directional, panel-level planning aid rather than a precise instrument for any single country's binding emissions target.

10. Reproducibility Information

10.1 Randomness

random_state=42 was fixed for Random Forest and XGBoost. KNN and SVR are deterministic given their inputs and hyperparameters, so no seed is required for them; the shared chronological train/validation/test split is itself fully deterministic (defined by year cutoffs, not random sampling), removing another common source of run-to-run variability.

10.2 Environment

The notebook was developed and executed in Google Colab. Recommended reproducibility artifacts to include alongside the submission: a requirements.txt (or environment.yml) pinning the versions of pandas, numpy, matplotlib, scikit-learn, and xgboost actually used in the Colab runtime at submission time.

10.3 Execution and file structure

The notebook runs top to bottom in four parts — Part 1 (data loading, audit, EDA), Part 2 (cleaning, feature engineering, splitting, baseline), Part 3 (model training, tuning, comparison, feature importance), and Part 4 (final test evaluation, diagnostics, refit, and forecast) — and depends on a single external input, owid-co2-data.csv, uploaded to the Colab session or read from a mounted Google Drive path. A submission-ready README should state this exact run order and the expected location of that CSV file, since the current notebook references a hard-coded Colab path (/content/owid-co2-data.csv) that should be replaced with a relative, documented path before final submission.

10.4 Artifacts

All figures and tables presented in this report (Sections 4, 7, 8, 9) are regenerated directly from cells in the submitted notebook and will reproduce identically on a clean re-run, given the same input CSV and package versions, because every stochastic component is seeded and every split boundary is a fixed year cutoff rather than a random draw.

11. Conclusion, Practical Interpretation, Responsible-Use Boundary, and Future Improvement

11.1 Conclusion

A linear-kernel Support Vector Regression model, trained on log-transformed CO2 with lagged historical features and a leakage-safe chronological split, forecasts national CO2 emissions three years ahead with a test R^2 of 0.93 and a mean absolute error of roughly 58 million tonnes — a large improvement over the mean baseline (negative R^2). The result supports the assessment brief's core principle: the simplest model that fit the data's largely linear, autoregressive structure outperformed more complex tree ensembles, and that outcome was reached through leakage checks, a proper baseline, and honest comparison rather than by choosing the most sophisticated available algorithm.

11.2 Practical interpretation

The model confirms that a country's own recent emissions trajectory and energy consumption are, by a wide margin, the strongest available predictors of its near-future emissions — a useful, if unsurprising, planning signal: absent a major policy or economic shock, emissions three years out are best anchored to the last one to three years of history rather than to slower-moving variables like GDP or population alone.

11.3 Responsible-use boundary

As stated in Section 9.5, the model is a panel-level directional planning aid. It should not be used as the sole basis for binding, country-specific emissions targets or climate-finance decisions, particularly for mid-sized economies

like Pakistan where test-set percentage errors were largest, or for any country outside the 17-country panel it was trained on.

11.4 Future improvement

- Extend the panel with country-specific random effects or hierarchical modelling so smaller economies are not dominated by India-scale patterns in a pooled fit.
- Replace the single train/validation/test cut with rolling-origin (walk-forward) cross-validation to better quantify how forecast uncertainty grows with horizon and to test robustness across multiple historical cutoffs.
- Incorporate exogenous shock indicators (e.g., global energy prices, major policy changes) to explain outlier years like the Cambodia 2020–2024 sustained increase identified in Section 4.4.
- Report SVR-native diagnostics (e.g., permutation importance computed directly on the winning model) rather than relying on Random Forest/XGBoost importances as a proxy, to more faithfully interpret the deployed model itself.

12. References and Contribution Statement

12.1 References

- Our World in Data. CO₂ and Greenhouse Gas Emissions dataset (owid-co2-data.csv). Compiled from the Global Carbon Project, the Energy Institute Statistical Review of World Energy, and UN/World Bank population and GDP series. Licensed under Creative Commons BY 4.0.
- Pedregosa, F. et al. Scikit-learn: Machine Learning in Python. Journal of Machine Learning Research, 12, 2825–2830.
- Chen, T. and Guestrin, C. XGBoost: A Scalable Tree Boosting System. Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, 2016.
- Course materials: Programming in Python — Final-Term Project Requirements, Summer 25-26 Semester.

12.2 Contribution statement

Member	Primary Contribution
[HAMIM IBNE RAHMAN]	[e.g., data acquisition, cleaning, and audit (Part 1)]
[SYED SHAFKAT RAIYAN]	[e.g., feature engineering, splitting, and baseline (Part 2)]
[MD ABU RABBIUL RIYAD]	[e.g., model training, tuning, and comparison (Part 3)]
[MOHSIN MAMUN]	[e.g., final evaluation, forecasting, and report writing (Part 4)]

All members reviewed the complete notebook and this report and can individually defend every stage of the workflow, per the demonstration and viva expectations in the course requirements.